

Business Case: Target SQL

Submitted by
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1. Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1.1) Data type of all columns in the "customers" table.

Ans:

```
select column_name, data_type
from Target.INFORMATION_SCHEMA.COLUMNS
where table_name = 'customers';
```

Row	column_name	data_type
1	customer_id	STRING
2	customer_unique_id	STRING
3	customer_zip_code_prefix	INT64
4	customer_city	STRING
5	customer_state	STRING

Insights:

- Here in the customer table of the target dataset we can see the column names and their datatypes besides them. We can see there are two different datatypes one is string and one is integer.

1.2) Get the time range between which the orders were placed.

Ans:

```
select min(order_purchase_timestamp) as first_purchase,
max(order_purchase_timestamp) as last_purchase
from `Target.orders`;
```

Row	first_purchase	last_purchase
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

Insights:

- The first order in the dataset was placed on September 4, 2016, and the last order on October 17, 2018. This gives a time span of just over 2 years during which the 100,000 orders were placed.

1.3) Count the Cities & States of customers who ordered during the given period.

Ans:

```
select count(distinct customer_city) as no_of_cities, count(distinct
customer_state) as no_of_states
from `Target.customers` c
join `Target.orders` o
on c.customer_id = o.customer_id
```

Row	no_of_cities	no_of_states
1	4119	27

Insights:

- Orders were placed by customers from **4,119 different cities** spread across **27 states** in Brazil.
- The large number of cities suggests that Target has a broad geographical reach within Brazil, making it accessible to customers in both urban and potentially rural areas.

Recommendation:

- Analyze the order distribution by city and state to identify key regions with the highest and lowest sales. This can help Target optimize its marketing and logistics in underperforming regions.

2) In-depth Exploration:

2.1) Is there a growing trend in the no. of orders placed over the past years?

Ans:

```
select
extract(year from order_purchase_timestamp) as year,
count(order_id) as no_of_orders
from `Target.orders`
group by year
order by year
```

Row	year	no_of_orders
1	2016	329
2	2017	45101
3	2018	54011

Insights:

- There is a significant growth in the number of orders placed between 2016 and 2018.

- In **2016**, only **329 orders** were placed, likely due to the platform or operation being in an early phase.
- In **2017**, orders jumped to **45,101**, indicating a major expansion or increased adoption by customers.
- By **2018**, orders further increased to **54,011**, showing continued growth with a **19.8% increase** from 2017 to 2018.

2.2) Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

Ans:

```
select extract(year from order_purchase_timestamp) as year,
extract(month from order_purchase_timestamp) as month,
count(order_id) as No_of_orders
from `Target.orders`
group by year, month
order by year, month;
```

JOB INFORMATION		RESULTS	CHART	JSON	EXECU
Row	year ▼	month ▼	No_of_orders ▼		
1	2016	9	4		
2	2016	10	324		
3	2016	12	1		
4	2017	1	800		
5	2017	2	1780		
6	2017	3	2682		
7	2017	4	2404		
8	2017	5	3700		
9	2017	6	3245		
10	2017	7	4026		

Insights:

- The dataset only includes orders from September to December. Notably, October 2016 saw 324 orders, while other months in 2016 had minimal activity (4 in September and 1 in December).
- In 2017, the number of orders steadily increases from January (800 orders) to a peak in November (7,544 orders), followed by a slight dip in December (5,673 orders).
- 2018 shows a similar trend, with high order volumes in the early months (e.g., January with 7,269 orders) and consistent order numbers above 6,000 in subsequent months.

November 2017 saw a substantial spike, likely due to promotional events such as **Black Friday** or other seasonal sales. This pattern suggests that November is a high-demand month for orders, possibly driven by discounts and marketing campaigns during holiday periods.

2.3) During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

- 0-6 hrs : Dawn
- 7-12 hrs : Mornings
- 13-18 hrs : Afternoon
- 19-23 hrs : Night

Ans:

```
with cte1 as
(select
case
when extract(hour from order_purchase_timestamp) between 0 and 6 then
'Dawn'
when extract(hour from order_purchase_timestamp) between 7 and 12 then
'Mornings'
when extract(hour from order_purchase_timestamp) between 13 and 18 then
'Afternoon'
when extract(hour from order_purchase_timestamp) between 19 and 23 then
'Night'
end as Timing,
order_id
from `Target.orders`)

select
Timing, count(order_id) as no_of_order
from cte1
group by Timing
order by no_of_order desc;
```

Query results			
JOB INFORMATION		RESULTS	CHART
Row	Timing	no_of_order	JSON
1	Afternoon	38135	
2	Night	28331	
3	Mornings	27733	
4	Dawn	5242	

Insights:

- The data shows that most orders are placed during the **Afternoon** (13-18 hrs), accounting for **38,135 orders**. This is the peak time for Brazilian customers, followed by **Night** (19-23 hrs) with **28,331 orders**.
- **Morning** (7-12 hrs) sees slightly fewer orders (**27,733 orders**) compared to the night.
- **Dawn** (0-6 hrs) has the least number of orders (**5,242 orders**), indicating it is the least active time for placing orders.

Recommendation:

- Since most customers place orders in the Afternoon and Night, Target should focus its marketing campaigns, promotions, and social media

engagement during these hours to maximize reach and sales. This could include time-specific offers or flash sales.

3. Evolution of E-commerce orders in the Brazil region:

3.1) Get the month on month no. of orders placed in each state.

Ans:

```
select
format_date('%Y-%m', order_purchase_timestamp) as year_month,
c.customer_state as state,
count(t.order_id) as no_of_orders
from `Target.orders` as t
join `Target.customers` as c
on t.customer_id = c.customer_id
group by year_month,state
order by year_month, no_of_orders desc;
```

Row	year_month	state	no_of_orders
1	2016-09	SP	2
2	2016-09	RR	1
3	2016-09	RS	1
4	2016-10	SP	113
5	2016-10	RJ	56
6	2016-10	MG	40
7	2016-10	RS	24
8	2016-10	PR	19
9	2016-10	SC	11
10	2016-10	GO	9

Insights:

- SP consistently has the highest number of orders placed each month, indicating its status as a crucial market for Target in Brazil. The strong performance in SP suggests that **Target's operations, including marketing, logistics, and customer engagement, are particularly effective in this state** compared to others.
- The concentration of orders in SP shows that a significant portion of Target's revenue may be driven by this one state.

Recommendation:

- Expand product offerings and marketing campaigns in SP to continue capitalizing on its strong demand. Focus on high-demand products and explore cross-selling opportunities to increase average order value.

3.2) How are the customers distributed across all the states?

Ans:

```
select
customer_state, count(customer_unique_id) as no_of_unique_customers
```

```
from `Target.customers`
group by customer_state
order by no_of_unique_customers;
```

Row	customer_state	no_of_unique_customers
1	RR	46
2	AP	68
3	AC	81
4	AM	148
5	RO	253
6	TO	280
7	SE	350
8	AL	413
9	RN	485
10	PI	495

Insights:

- **São Paulo (SP) has the highest number of customers, followed by Rio de Janeiro (RJ) and Minas Gerais (MG).** This suggests that the bulk of Target's customer base is located in Brazil's most populous and economically strong states. These regions are critical for driving Target's overall sales performance.
- States like **Roraima (RR)**, with the lowest number of customers, represent regions with low population density, weaker economies, or limited brand presence for Target.

Recommendation:

- Continue strengthening customer engagement and loyalty programs in high demand states, as they are the largest markets. This could include region-specific promotions, loyalty rewards, and faster delivery options to maintain strong customer relationships.
- Invest in targeted marketing and outreach efforts in lower-demand states like RR. Focus on building brand awareness and improving customer accessibility through digital marketing, social media, and local partnerships.

4. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

4.1) Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).

Ans:

```
with pay as
```

```
(select
```

```
tbl1.month,
```

```

tbl1.month_name,
round(sum(if(year = 2018,payment,0)),2) as payment_2018,
round(sum(if(year = 2017,payment,0)),2) as payment_2017
from
(select
extract(year from o.order_purchase_timestamp) as year,
extract(month from o.order_purchase_timestamp) as month,
format_datetime('%B', order_purchase_timestamp) as month_name,
sum(p.payment_value) as payment
from `Target.orders` as o
join `Target.payments` as p on o.order_id = p.order_id
group by year, month, month_name) as tbl1
where tbl1.month between 01 and 08
group by tbl1.month_name, tbl1.month)

select
pay.month, pay.month_name, pay.payment_2018, pay.payment_2017,
concat(round(((pay.payment_2018-
pay.payment_2017)/pay.payment_2017)*100,2),'%') as cost_increase_per
from pay
order by pay.month

```

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS		EXECUTION GRAPH
Row	month	month_name		payment_2018	payment_2017	cost_increase_per	
1	1	January		1115004.18	138488.04	705.13%	
2	2	February		992463.34	291908.01	239.99%	
3	3	March		1159652.12	449863.6	157.78%	
4	4	April		1160785.48	417788.03	177.84%	
5	5	May		1153982.15	592918.82	94.63%	
6	6	June		1023880.5	511276.38	100.26%	
7	7	July		1066540.75	592382.92	80.04%	
8	8	August		1022425.32	674396.32	51.61%	

Insights:

- The highest percentage increase in the cost of orders occurred in **January**, followed by **February** and **April**. This suggests that at the beginning of 2018, Target experienced a surge in order costs compared to the same period in 2017. This growth could be due to several factors: **higher demand post-holiday season, successful promotional campaigns**

- The drop-off in growth during August may point to **seasonal trends** or a lack of significant promotions or events compared to the earlier months.

Recommendation:

- Maximize promotions in high-growth months like January, February, and April by offering exclusive deals or bundling popular products. Ensuring adequate inventory and optimizing logistics during these months can help capture more revenue from the growing demand.
- **Implement new promotional strategies in August** to drive up sales and counteract the slowdown in cost growth.

4.2) Calculate the Total & Average value of order price for each state.

Ans:

```
select
c.customer_state,
round(sum(p.payment_value),2) as total_amount,
round(avg(p.payment_value),2) as avg_amount
from `Target.customers` c
join `Target.orders` o on c.customer_id = o.customer_id
join `Target.payments` p on p.order_id = o.order_id
group by c.customer_state
order by total_amount desc, avg_amount desc;
```

Row	customer_state	total_amount	avg_amount
1	SP	5998226.96	137.5
2	RJ	2144379.69	158.53
3	MG	1872257.26	154.71
4	RS	890898.54	157.18
5	PR	811156.38	154.15
6	SC	623086.43	165.98
7	BA	616645.82	170.82
8	DF	355141.08	161.13
9	GO	350092.31	165.76
10	ES	325967.55	154.71

Insights:

- **SP has the highest total order value**, followed by Rio de Janeiro (RJ) and Minas Gerais (MG). This indicates the significance of SP as the most important market for Target in Brazil, likely driven by its large customer base, economic strength, and higher purchasing power.

Recommendation:

- Focus on further optimizing marketing and sales strategies in SP, RJ, and MG as these states are already generating the highest total order values.

Consider expanding premium product offerings or launching exclusive deals that cater to the preferences of customers in these regions.

4.3) Calculate the Total & Average value of order freight for each state.

Ans:

```
select
c.customer_state as state_code,
round(sum(oi.freight_value),2) as Total_Freight_cost,
round(avg(oi.freight_value),2) as Avg_Freight_cost
from `Target.customers` c
join `Target.orders` o on o.customer_id = c.customer_id
join `Target.order_items` oi on oi.order_id = o.order_id
group by c.customer_state
order by Total_Freight_cost desc, Avg_Freight_cost desc
```

Row	state_code	Total_Freight_cost	Avg_Freight_cost
1	SP	718723.07	15.15
2	RJ	305589.31	20.96
3	MG	270853.46	20.63
4	RS	135522.74	21.74
5	PR	117851.68	20.53
6	BA	100156.68	26.36
7	SC	89660.26	21.47
8	PE	59449.66	32.92
9	GO	53114.98	22.77
10	DF	50625.5	21.04

Insight:

- **SP has the highest total freight value**, followed by RJ and MG. This is consistent with SP's position as the state with the highest total order value, indicating that **high order volumes drive higher shipping costs**.
- The high total freight value in SP reflects a **concentration of demand in this region**, where both order size and frequency likely contribute to greater logistics expenses.

Recommendation:

- Investigate logistics efficiency in SP, RJ, and MG to identify opportunities for reducing freight costs. This could include renegotiating contracts with

shipping providers, improving warehouse proximity to customer hubs, or offering more economical delivery options.

5. Analysis based on sales, freight and delivery time.

5.1) Find the no. of days taken to deliver each order from the order's purchase date as delivery time.

Also, calculate the difference (in days) between the estimated & actual delivery date of an order.

Do this in a single query.

Ans:

```
select  
  
timestamp_diff(order_delivered_customer_date, order_purchase_timestamp,  
Day) as time_to_deliver,  
  
timestamp_diff(order_estimated_delivery_date, order_delivered_customer_date,  
Day) as diff_estimated_delivery  
  
from `Target.orders`  
  
where lower(order_status) = 'delivered'  
  
order by time_to_deliver, diff_estimated_delivery;
```

Row	time_to_deliver	diff_estimated_delivery
1	209	-181
2	208	-188
3	195	-165
4	194	-155
5	194	-161
6	194	-166
7	191	-175
8	189	-167
9	188	-159
10	187	-144

Insights:

- The longest delivery time of 209 days suggests that certain orders experience extreme delays. **Prolonged delivery times** like 209, 208, 195, and 194 days indicate inefficiencies or issues in the logistics or fulfillment process, which can negatively impact customer satisfaction and trust in the service.

Recommendation:

- Conduct a detailed analysis of orders with extremely long delivery times (209, 208, 195 days) to understand the root causes. Are these orders from remote areas? Were there stock shortages or issues with fulfillment

centers? Identify patterns and take corrective action to avoid such delays in the future.

5.2) Find out the top 5 states with the highest & lowest average freight value.

Ans:

```
(select
c.customer_state as State,
round(avg(oi.freight_value),2) as Avg_freight
from `Target.order_items` oi
join `Target.orders` o on oi.order_id = o.order_id
join `Target.customers` c on c.customer_id = o.customer_id
group by c.customer_state
order by Avg_Freight desc
limit 5)
union all
(select
c.customer_state as State,
round(avg(oi.freight_value),2) as Avg_freight
from `Target.order_items` oi
join `Target.orders` o on oi.order_id = o.order_id
join `Target.customers` c on c.customer_id = o.customer_id
group by c.customer_state
order by Avg_Freight
limit 5);
```

Row	State	Avg_freight
1	RR	42.98
2	PB	42.72
3	RO	41.07
4	AC	40.07
5	PI	39.15
6	SP	15.15
7	PR	20.53
8	MG	20.63
9	RJ	20.96
10	DF	21.04

Insights:

- **Roraima (RR), Paraíba (PB), Rondônia (RO), Acre (AC), and Piauí (PI)**, which have the highest average freight values, are largely remote or less urbanized areas. These states tend to have higher shipping costs due to their distance from major distribution hubs, as well as less developed infrastructure and fewer logistical options.
- The states with the lowest average freight values, **São Paulo (SP), Paraná (PR), Minas Gerais (MG), Rio de Janeiro (RJ), and Distrito Federal (DF)** are major urban centers with **better infrastructure, higher population density, and closer proximity to fulfillment centers**.

Recommendation:

- Investigate partnerships with local logistics providers or alternative shipping methods (e.g., regional couriers) in remote states like RR, PB, RO, AC, and PI to reduce freight costs and delivery times.

5.3) Find out the top 5 states with the highest & lowest average delivery time.

Ans:

```
(select
c.customer_state as state,

avg(timestamp_diff(order_delivered_customer_date, order_purchase_timestamp,
day)) as avg_time_to_deliver

from `Target.customers` c

join `Target.orders` o on c.customer_id = o.customer_id

group by c.customer_state

order by avg_time_to_deliver desc

limit 5)

union all

(select
c.customer_state as state,

avg(timestamp_diff(order_delivered_customer_date, order_purchase_timestamp,
day)) as avg_time_to_deliver

from `Target.customers` c

join `Target.orders` o on c.customer_id = o.customer_id

group by c.customer_state

order by avg_time_to_deliver

limit 5);
```

Row	state	avg_time_to_deliver
1	RR	28.97560975609...
2	AP	26.73134328358...
3	AM	25.98620689655...
4	AL	24.04030226700...
5	PA	23.31606765327...
6	SP	8.298061489072...
7	PR	11.52671135486...
8	MG	11.54381329810...
9	DF	12.50913461538...
10	SC	14.47956019171...

Insights:

- The states with the highest average delivery times—**Roraima (RR)**, **Amapá (AP)**, **Amazonas (AM)**, **Alagoas (AL)**, and **Pará (PA)**—are predominantly located in remote or northern regions of Brazil. These areas face logistical challenges such as difficult terrain, less developed infrastructure, and greater distances from major distribution centers, which contribute to longer delivery times.
- The states with the lowest average delivery times—**São Paulo (SP)**, **Paraná (PR)**, **Minas Gerais (MG)**, **Distrito Federal (DF)**, and **Santa Catarina (SC)**—are all urbanized and economically significant regions with **well-developed infrastructure** and closer proximity to major fulfillment centers.

Recommendation:

- **Analyze current logistics routes and providers** serving remote regions like RR, AP, AM, AL, and PA to identify potential inefficiencies. Consider **exploring alternative transportation methods** or working with local delivery services that are more familiar with these areas.

5.4) Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

Ans:

```
select
c.customer_state as state,
round(avg(date_diff(order_estimated_delivery_date,order_delivered_customer_date, Day))) as avg_delivery_time,
from `Target.customers` c
join `Target.orders` o on c.customer_id = o.customer_id
where lower(order_status) = 'delivered'
group by c.customer_state
order by avg_delivery_time desc
```

limit 5;

Row	state	avg_delivery_time
1	AC	20.0
2	AP	19.0
3	AM	19.0
4	RO	19.0
5	RR	16.0

Insights:

- The top 5 states where the order delivery is really fast are **AC, AP, AM, RO, RR** – indicating a higher efficiency in logistics and delivery operations in these areas.

6. Analysis based on the payments:

6.1) Find the month on month no. of orders placed using different payment types.

Ans:

```
select
format_date('%Y-%m', o.order_purchase_timestamp) as month_year,
p.payment_type, count(o.order_id) as No_of_order
from `Target.orders` o
join `Target.payments` p on o.order_id = p.order_id
where p.payment_type is not null
group by 1,2
order by month_year;
```

Row	month_year	payment_type	No_of_order
1	2016-09	credit_card	3
2	2016-10	credit_card	254
3	2016-10	UPI	63
4	2016-10	voucher	23
5	2016-10	debit_card	2
6	2016-12	credit_card	1
7	2017-01	credit_card	583
8	2017-01	UPI	197
9	2017-01	voucher	61
10	2017-01	debit_card	9

Insights:

- From the above statement we can tell that most of the people prefer to pay with credit card followed by upi

Recommendation:

- The ecommerce company should offer exclusive benefits and rewards to customers who use credit cards as their payment method such as,
 - Cashback or discount offers.
 - Early access to sales and promotions.
 - Free shipping or express delivery.
- The company should offer reward points to customers who use UPI as their payment method to give reward points for their next purchase etc.

6.2) Find the no. of orders placed on the basis of the payment installments that have been paid.

Ans:

```
select
p.payment_installments as no_of_installment,
count(p.order_id) as no_of_orders
from `Target.payments` p
group by no_of_installment
order by no_of_orders desc;
```

Row	no_of_installment	no_of_orders
1	1	52546
2	2	12413
3	3	10461
4	4	7098
5	10	5328
6	5	5239
7	8	4268
8	6	3920
9	7	1626
10	9	644

Insights:

- Most of the people selects to pay in one single installment, while little no of people likes to pay by 24 installments.

Recommendation:

- Optimize and promote single-payment options for the majority of customers while enhancing flexible installment plan (such as upto 24 months) to cater the higher value purchase to ensure a balanced and customer-centric payment strategy.

